questions in this exam.

Multiple choice Questions (Circle **only ONE answer per question** on the external answer sheet. If more than one answer is selected, the question will NOT be graded.) [20 marks]

1. After performing the following operation Dequeue (Q, X), the front element of the queue will be
(a) ~~X~~ (b) ~~Q~~ (c) Null (d) **Invalid Function**

(a) The top of the Stack will equal to Max **(b) The top of the Stack will equal to Max-1** **(c) The top of the Stack will equal to 0** **(d) Nothing will happen for the top**

2. After performing the following operations IsFull (S) on a stack

3. ______ is a set of data values and associated operations that are specified accurately, independent of any particular implementation. **(a) Stack** **(b) ~~Tree~~** **(c) Abstract Data Type** **(d) ~~Grap~~**

ADT

4. Which of the following is a primitive data structure? **(a) Boolean (b) Integer (c) Character (d) all of mentioned**

5. Which of the following is a non linear data structure? **(a) ~~array~~ (b) ~~stack~~ (c) Tree (d) ~~Linked~~ list**

6. Which data structure is most commonly used to convert a recursive algorithm into a non-recursive (iterative) form? **A. Stack B. Queue C. Graph D. Array**

7. In general, List data structure allow: **(A) Insertions and removals anywhere. (B) Insertions and removals ~~only at~~ one end. (C) Insertions at the back and removals from the front. (D) ~~None~~ of the above.**

8. Choose the correct option for the following declaration and statements are the following:

(1)L1= &z; (2)p =L1; (3)Free(y); (4)z = (Node *) malloc(sizeof(Node));

(5)q = *x; (6)p = r; (7)L1=Null

Typedef struct node { char c; Node *next Node	Typedef Node *List Node *p, *q, *r; Node x, y, z; List L

A. Statements 1, 3, 5, and 6 are valid **B. Statements 2, 3, 4, and 5 are valid**
C. Statements 1, 2, 6, and 7 are valid **D. Statements 4, 5, 6, and 7 are valid**

9. In a stack, inserting (push) an element takes: **A) O(1) B) O(log n) C) O(n) D) O(n^2)**

10. Consider the situation where no data structures like arrays or linked lists are available to you. How many stacks are needed to implement a queue? **A.** 1 **B.** 2 C. 3 D. Cannot be implemented without arrays or linked lists

stack => Arr

11. In an array-based list, what is the time complexity of inserting a new element? ~~A)~~ O(1) **B) ~~O(log n)~~**
C) O(n) **D) ~~O(n^2)~~**

12. What are the preconditions for the function AddLists(Stack L1, Stack L2, Stack *L3) which adds the elements of L1 and L2 into L3? **A)** L1 and L2 should be empty and L3 should ~~be full~~ **B.**L1 and L2 should be non-empty and L3 should be empty **C.** L1 and L2 should be sorted and L3 should be initialized **D.**L1 and L2 should be NULL and L3 should ~~be NULL~~

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13. What is the purpose of the (newFunc)function? **A.** It pushes a new element onto the stack recursively. **B.** It deletes only the top element of the stack. **C.** It frees all nodes in the stack recursively. **D.** It causes a segmentation fault due to incorrect pointer usage.

14. What is the main issue in the (F) function? **A.** The function causes memory leakage because it doesn't free the nodes. **B.** The line *l1 = *l1->next; will cause a segmentation fault because *l1 is already set to NULL. **C.** The loop condition is incorrect and will cause an infinite loop. **D.** The function correctly deletes the entire list without errors.

15. Which of the following should be inserted in the following code to retrieve the **first element** in the queue?

void FirstElement(EntryType *pe, Queue *pq){ ....; }

a) *pq = pe->front->entry; **b)** *pe = pq->entry; **c)** *pe = pq->front->entry; **d)** *pq = pe->entry;

16. We can prevent an array-based stack data structure from containing **unsorted elements** by modifying the function:

a) Pop() b) Push() c) CreateStack() d) IsEmpty()

17. If data is a circular array of CAPACITY elements, and rear is an index into that array, what is the formula for the index after rear? **A)** (rear % 1) + CAPACITY **B)**rear % (1 + CAPACITY) **C)** (rear + 1) % CAPACITY **D)** rear + (1 % CAPACITY)

18. A stack S can have a maximum of ten elements. If the values **K**, **L**, **M**, and **O** are pushed into it, then the top will be equal to **4**. After executing IsEmpty(s), the top will be: ....4.........., and after executing CreateStack(&s), the top will be: ....0........

a) 4, -1 b) 0, -1 c) -1, 0 d) 0, 4

19. What is the wrong of the following code? void main(){ Stack s; Push(&s, 1); create Stack(&s)}

a) The Push function is called before the stack is created. b) Push cannot be used with integers.
c) &s cannot be passed to a function. d) The stack must be empty before pushing.

20. What is the main advantage of a array over an linked list?

A) Faster access to elements **B) All of mentioned**

C) Less memory usage **D) Easier indexing**

```
void newFunc (Stack *sp){
  if (sp->top){
    Node *np = sp->top;
    np = np->next;
    free(sp->top);
    sp->top = np;
    newFunc(sp); } }
```

```
void F(List *l1){
  while(*l1 != NULL){
    *l1 = NULL;
    *l1 = *l1->next; } }
```